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M1 MACRO

Group A

Session 8

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Intertemporal budget constraint
(slanting in t)

$$A_{t+1} + C_t = (1+r)A_t + Y_t$$

~~$$A_{t+1} = (1+r)A_t + Y_t - C_t$$~~

$$A_t = \frac{1}{1+r} (A_{t+1} + C_t - Y_t) \quad \text{in } t$$

$$\begin{aligned} A_{t+1} &= \frac{1}{1+r} (A_{t+2} + C_{t+1} - Y_{t+1}) \quad \text{in } t+1 \\ &= \frac{1}{1+r} \left(\left[\frac{1}{1+r} (A_{t+3} + C_{t+2} - Y_{t+2}) \right] + C_{t+1} - Y_{t+1} \right) \end{aligned}$$

$$A_{t+1} = \left(\frac{1}{1+r}\right)^2 A_{t+3} + \left(\frac{1}{1+r}\right)^2 (C_{t+2} - Y_{t+2}) + \frac{1}{1+r} (C_{t+1} - Y_{t+1})$$

$$\underbrace{\quad}_{\frac{1}{1+r} (A_{t+1} + C_{t+3} - Y_{t+3})}$$

⋮

$A_T < 0$
Ponzi scheme

$$A_t = \left(\frac{1}{1+r}\right)^T A_{t+T+1} + \frac{1}{1+r} \sum_{j=0}^T \left(\frac{1}{1+r}\right)^j (C_{t+j} - Y_{t+j})$$

We impose

$$\lim_{T \rightarrow +\infty} \left(\frac{1}{1+r}\right)^T A_{t+T+1} = 0 = \begin{cases} \text{TRANSVERSALITY} \\ \text{CONDITION} \end{cases} \quad (= \text{condition for optimality})$$

with the T.C. and rewriting as of period t :

$$A_t = \frac{1}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j (C_{t+j} - Y_{t+j})$$

or

$$A_t + \sum_{j=0}^{\infty} \frac{Y_{t+j}}{(1+r)^j} = \sum_{j=0}^{\infty} \frac{C_{t+j}}{(1+r)^j}$$

• $C_{t+j} - Y_{t+j}$: can't be always > 0 ?

Human Wealth

$$H_t = \frac{1}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j y_{t+j}$$

Assume $y_{t+j} = \bar{y} \forall j$

Then $H_t = \frac{1}{1+r} \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j \bar{y}$

$$= \bar{y} \times \frac{1}{1+r} \times \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j$$

$$H_t = \bar{y} \frac{1}{1+r} \times \left(\frac{1}{1 - \frac{1}{1+r}} \right) = \bar{y} \frac{1}{1+r} \frac{1+r}{r} = \frac{\bar{y}}{r}$$

Undetermined coefficients

$$C_T = \phi_1 A_t + \phi_2 \overline{Y}$$

. $A_{t+1} = A_t$ from Euler $= \rho A_t + \overline{Y}$

. BC $A_{t+1} = (1+\rho)A_t + \overline{Y} - \underbrace{C_t}_{\phi_1 A_t + \phi_2 \overline{Y}}$
 $= A_t$

$$\Leftrightarrow A_t = (1+\rho)A_t + \overline{Y} - \phi_1 A_t - \phi_2 \overline{Y}$$

$$\Leftrightarrow (\underbrace{\rho - \phi_1}_{=0}) A_t + (\underbrace{1 - \phi_2}_{=0}) \overline{Y} = 0$$

$\Rightarrow \phi_1 = \rho, \phi_2 = 1$

$\begin{matrix} \cancel{A_t} \\ \cancel{\overline{Y}} \end{matrix}$

Rational Expectations

• Assume I need to forecast X_{t+1}

• I need to form $E_t X_{t+1}$

• Expectation error $\varepsilon_{t+1}^x = X_{t+1} - E_t X_{t+1}$

Rational expect $\Rightarrow \varepsilon^x$ has some remarkable properties

$$(1) \quad E_t \varepsilon_{t+1}^x = 0$$

$$(2) \quad \text{cov}(\varepsilon_{t+1}^y, \varepsilon_t^x) = 0$$

①

$$\varepsilon^X_{t+1} = X_{t+1} - E_t X_{t+1}$$

$$E_t \varepsilon^X_{t+1} = E_t [X_{t+1} - E_t X_{t+1}]$$

$$= E_t X_{t+1} - E_t (E_t X_{t+1})$$

$$= E_t X_{t+1} - E_t X_{t+1} = 0$$

I am not systematically wrong.

$$(2) \text{Cov}(\varepsilon_t^x, \varepsilon_{t+1}^x) = 0$$

$$\hookrightarrow E_t(X_t X_{t+1}) = E_t(X_t) E_t(X_{t+1})$$

$$= 0$$